October 21, 2026

The Editor-in-Chief
Computer Modeling in Engineering & Sciences (CMES)
Tech Science Press

Dear Editor,

Re: Submission of a new manuscript for consideration in CMES

We are pleased to submit our manuscript entitled *"Fixed Points and Certified Boundedness of Unrolled Born/Contrast-Source Inverse Scattering Networks"* for consideration as a Research Article in *Computer Modeling in Engineering & Sciences*. The corresponding author is Longgang Wang (Luoyang Institute of Science and Technology, Luoyang 471000, Henan, China; email: longgangwang@163.com; ORCID: 0009-0008-1394-7953).

Originality and author approval

This manuscript is original and has not been published previously. It is not under consideration for publication elsewhere, in whole or in part. All authors have read the final manuscript and approved its submission to the journal. They accept full responsibility for the manuscript's delivery and contents. The work reports no human or animal subjects. There are no copyright or disclosure issues associated with the manuscript. All authors declare that they have no conflicts of interest regarding the submitted manuscript.

What the paper contributes

Deep unrolling of classical iterative solvers is a leading design principle for electromagnetic inverse scattering, and unrolled architectures reported in the literature outperform their classical ancestors empirically. What has been missing is a layer-level account of *why* such a network is stable, when it is provably not, and what can be certified about it. This paper supplies that account. Its four contributions, which mirror the manuscript's contribution list, are:

1. **Layer-level operator form.** The discrete Lippmann–Schwinger system is written as a fixed-point iteration of a physics operator T^{phys} on the state $z = (\chi, w)$; an unrolled network is exactly a cascade of layers $T_k = (1-\theta_k)I + \theta_k T^{\mathrm{phys}}_{\alpha_k,\gamma_k} + H_k$ with H_k a learned correction, and the fixed points of the trained map are characterized exactly.

2. **Impossibility of global contraction, in threshold form.** For the faithful Born/contrast-source composite, the bilinear coupling $\chi \odot E$ makes one block of the layer Jacobian *exactly* $\theta\alpha\,\mathrm{diag}(E)$. Because that block belongs to the second block row, which the learned correction does not touch, the bound $\sigma_{\max}(DT_k) \geq \theta\alpha\mathcal{E} > 1$ holds *for every correction and every Lipschitz constant* above an explicit step-size threshold. Global Euclidean contraction is therefore unattainable on the measured step-size grid, and stability guarantees must be layered rather than global.

3. **Layer-wise certification instead of per-layer contraction.** We prove what *is* available: an a-priori field-subsystem contraction ($\beta_0 = \chi_{\max}\|\mathbf{G}_b\|_2 < 1$) that confines the rollout to an explicit compact set at arbitrary depth under a nonexpansive physics-prior projection, so infinite-depth extrapolation cannot diverge; an exact finite-depth noise accounting, including the non-normal factor $\kappa_2(V)$ that a naive $\Lambda/(1-\rho)$ omits; and an explicit constructive bound on the data-Lipschitz constant. The geometric bounds hold only *conditionally*, and we demonstrate on our own measurements where their hypotheses fail. The certificates are recomputed at the operator level on the published Institut Fresnel geometry, where the weak-scattering condition fails while

the non-contractivity bound tightens, and a reconstruction from measured fields is reported for nine phantom-frequency pairs without the trained weights.

4. **The price of certification, measured.** Pre-registered ablations over 800 unrolled layers isolate which ingredient delivers which guarantee: removing the projection from frozen weights makes the unconstrained network explode, and retraining without it restores empirical boundedness but pushes the measured contrast bound past the weak-scattering threshold, so no a-priori certificate survives. The amplitude budget buys the a-priori constants at a measurable accuracy cost, and bounding the correction trunk's input is what keeps the state inside the admissible box.

The paper also reports a per-layer cost audit and an identical-seed reproducibility floor, so that each numerical claim can be traced to a script and an archived artifact.

Why this journal

CMES publishes research of the highest quality in computational modeling and engineering sciences. Our contribution is squarely within its scope: methodologically, it combines operator theory of the discrete Lippmann–Schwinger formulation with contraction-mapping and spectral analysis to obtain fixed-point and boundedness certificates for unrolled inverse-scattering networks, and the certifying constants are computed on the 2-D TM method-of-moments scattering operator, an inverse/imaging problem in the journal's stated application domain. The framework is a reusable template that separates what can be certified from what is only measured.

Data and code availability

All numerical instruments, experiment drivers, audit scripts, and a frozen snapshot with a standard `md5sum -c` manifest are listed in the manuscript's Data and Code Availability section. A public archive URL or persistent identifier will accompany the final version.

Suggested reviewers

We are glad to suggest potential reviewers for this manuscript upon request.

We thank you for considering our manuscript and look forward to your response.

Sincerely,

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